

Executive Summary

PS63 should evolve into a **next-generation polar research data platform** with AI and advanced UX to unify data, publications, and people. This report proposes **12–15 novel features** beyond the initial 31, prioritised by impact and feasibility (High/Medium/Low). For each feature we detail its purpose, user value, data model (schema), key APIs, UI components, implementation complexity (S/M/L), and risks with mitigations. We also outline a **12–18 month roadmap** (by quarter) with deliverables, dependencies and resource estimates (Dev, Data, ML, QA, Infra), a competitor/benchmark analysis (Zenodo, Dryad, PANGAEA, NOAA, NASA Earthdata, ResearchGate) with feature gaps, and a recommended tech stack (frontend, backend, DB, search/semantic, ML, mapping, viz, hosting) with rationale. We include **privacy, security and data governance checklist** and a **source-grounded AI verification workflow**. Finally, we provide sample developer prompts (product brief & technical implementation) and illustrative Mermaid charts (roadmap Gantt chart and a knowledge-graph schema).

This analysis is based on recent literature on data platforms, semantic search and AI tools, competitor docs, and best practices for research data infrastructure [89†L236-L244] [65†L122-L130] .

Novel Features (beyond original 31)

1. **Real-time Data Streams & Alerting (Priority: High)** – Ingest live sensor, satellite or IoT data (e.g. polar weather stations, sea buoys) into PS63. Value: Users get up-to-date observations (e.g. temperature, ice coverage) and configurable **alerts** for events (e.g. sudden warming) to trigger research actions.
2. *Technical Feasibility:* Use APIs or MQTT for sensors; implement streaming ingestion pipelines.
3. *Complexity:* M (data infrastructure, streaming)
4. *Risk:* Data quality/noise; Mitigation: validate/flag anomalous data and provide raw vs cleaned toggles.
5. *Data Model:* **SensorData** entries with fields (timestamp, lat, lon, sensor_id, value, unit, quality_flag).
6. *APIs:* POST /api/streaming/sensordata (ingest), GET /api/sensordata?start=&end=&bbox= (query range).
7. *UI:* Map and chart widgets (e.g. live updating line chart, interactive map overlay), alert subscriptions page.
8. **Federated Data Search & Integration (Priority: High)** – Enable unified search across external repositories (e.g. NOAA, NASA, Zenodo, Dryad) via APIs. Value: PS63 becomes a one-stop portal; users find polar-relevant datasets/publications even if hosted elsewhere.
9. *Technical Feasibility:* Use external APIs (e.g. NASA CMR, Zenodo REST API [8†L281-L290]) and harvest metadata.
10. *Complexity:* M (requires integration, ETL or federated query)

11. *Risk*: API changes/outages; Mitigation: cache results, use OAI-PMH where available, monitor endpoints.
12. *Data Model*: **ExternalRecord** linking (`source_name` , `external_id` , `title` , `type` , `url` , `metadata`).
13. *APIs*: `GET /api/federatedsearch?q=...` (queries multiple sources and merges results).
14. *UI*: Consolidated search results page with indicators (e.g. "from NASA", "from Zenodo").

15. **Collaborative Analysis Workspaces (Priority: High)** – Provide cloud-hosted **Jupyter notebooks** or **RStudio** where researchers can code/analyse PS63 data without leaving the platform. Value: Eases data exploration and reproducibility (code lives with data) [89†L236-L244] .

16. *Technical Feasibility*: Containerized notebooks (e.g. JupyterHub) with pre-installed libraries; integrate with PS63 auth and storage.

17. *Complexity*: L (well-known systems exist)
18. *Risk*: Resource cost (compute/ram); Mitigation: quota per user, auto-stop idle sessions.
19. *Data Model*: **AnalysisSession** with `session_id` , `user_id` , `start_time` , `status` , `notebook_path` .
20. *APIs*: `POST /api/workspace/create` , `GET /api/workspace/{id}/files` , `POST /api/workspace/{id}/execute` .
21. *UI*: "My Workspace" page listing active sessions, file browser, "New Notebook" button.

22. **AI-driven Recommendations & Personalisation (Priority: High)** – Use ML to recommend relevant content (datasets, papers, people) based on user profile and activity. Value: Helps users discover related research (e.g. "Researchers who viewed this dataset also viewed...").

23. *Technical Feasibility*: Collaborative filtering or content-based recommender using Elasticsearch/KNN on embeddings.

24. *Complexity*: M (requires ML models/training)
25. *Risk*: Filter bubble, irrelevant recs; Mitigation: allow user feedback (thumbs up/down), periodically retrain models.
26. *Data Model*: Implicit (views, downloads) and explicit (likes) **Interaction** events with (`user_id` , `item_id` , `action` , `timestamp`).
27. *APIs*: `GET /api/recommendations?user={id}` returns top-N items.
28. *UI*: "Recommended for you" widget on dashboard; inline suggestions on item pages.

29. **3D & VR Geospatial Visualisations (Priority: Medium)** – Enable immersive mapping (e.g. **CesiumJS** globe or WebXR) of polar regions, expeditions and data. Value: Researchers and public can intuitively explore terrain, ice thickness layers, or expedition tracks in 3D [24†L6-L10] .

30. *Technical Feasibility*: Use CesiumJS (open 3D globe) or Mapbox GL; AR/VR support on mobile/VR headsets optional.

31. *Complexity*: L (many libraries)
32. *Risk*: Performance on low-end devices; Mitigation: provide 2D fallback.
33. *Data Model*: Tie **Location** data to 3D coordinates; e.g. **3DPoint** (`lat` , `lon` , `alt` , `description`).

34. *APIs*: `GET /api/locations3d?expedition=ID` to fetch geojson for rendering.
35. *UI*: “3D Map” view button on location or expedition pages; VR/AR mode toggle.
36. **Data Provenance & Lineage Tracking (Priority: Medium)** – Track and display data provenance (source of each datum, processing steps). Value: Enhances trust and reusability – users see the full chain (raw data → processing code → final result).
37. *Technical Feasibility*: Maintain metadata logs or RDF triples; use W3C PROV model if needed.
38. *Complexity*: M (schema design and logging)
39. *Risk*: Data overhead; Mitigation: compress logs, show summary visuals.
40. *Data Model*: **ProvenanceRecord**: (data_id, source_id, transformation_id, timestamp, actor).
41. *APIs*: `GET /api/provenance?data={id}` returns lineage.
42. *UI*: Expandable lineage diagram on data and dataset pages (e.g. source ↔ transformation ↔ usage).
43. **Automated Metadata Curation (Priority: Medium)** – Use AI/NLP to suggest tags, categories or fill missing metadata (e.g. extract location names from descriptions, classify research areas). Value: Improves data discoverability and quality with less manual effort.
44. *Technical Feasibility*: Use ML APIs or libraries (spaCy, HuggingFace NER) to auto-extract keywords, geotag from text.
45. *Complexity*: M
46. *Risk*: Incorrect tags; Mitigation: show suggestions for user approval.
47. *Data Model*: Extend existing item metadata with auto-generated **suggested_tags** list (string).
48. *APIs*: `POST /api/curate/metadata?item={id}` triggers ML analysis.
49. *UI*: On item edit page: “AI-suggested tags” checkbox list to accept/modify.
50. **Social Q&A and Discussion Forum (Priority: Medium)** – Add a community forum or comments thread on datasets, papers or expeditions. Value: Enables collaboration and peer support (“How was this sample collected?”, “Tips for analysis?”) and builds community.
51. *Technical Feasibility*: Use existing forum software (Discourse) or build comment system.
52. *Complexity*: S (moderation needed)
53. *Risk*: Spam or misinformation; Mitigation: require login, implement moderation tools, upvote system.
54. *Data Model*: **ForumPost** (post_id, parent_id, user_id, content, timestamp), **Thread** (thread_id, topic_id).
55. *APIs*: `GET /api/threads?item={id}`, `POST /api/thread/{id}/reply`.
56. *UI*: “Discussion” tab on each item page, threaded view with markdown support.
57. **Multi-modal Search (Priority: Medium)** – Enable searching by image or audio content. E.g., reverse-search by a photo of ice structure, or speech-to-text search within video transcripts.

Value: Makes media in PS63 searchable (e.g. find all images with “penguins” or all videos mentioning “ice thickness”).

58. *Technical Feasibility*: Use ML services: vision APIs (object recognition) and speech-to-text (for videos).

59. *Complexity*: M (needs ML pipelines)

60. *Risk*: High compute cost; Mitigation: pre-process on upload, index metadata.

61. *Data Model*: Store extracted **MediaLabels** (media_id, label, confidence) and **Transcription** (video_id, start_time, text).

62. *APIs*: GET /api/search/images?q=object , GET /api/search/videos?q=text integrating these indexes.

63. *UI*: Add “Images” and “Videos” search filters, with thumbnails and time-stamped highlights.

64. **Citizen Science Project Integration (Priority: Medium)** – Host or link to citizen-science tasks (e.g. animal-spotting on ice imagery, crowd-classify sea-ice types). Value: Engages public, crowdsources data annotation and extends outreach.

- *Technical Feasibility*: Integrate with platforms like Zooniverse or build mini-apps.
- *Complexity*: M
- *Risk*: Data quality of public contributions; Mitigation: consensus / expert review of contributions.
- *Data Model*: **CitizenTask** (task_id, description, linked_media, reward_points) and **Contribution** (user_id, task_id, answer, timestamp).
- *APIs*: GET /api/citizen/tasks , POST /api/citizen/tasks/{id}/submit .
- *UI*: “Participate” section with ongoing projects, leaderboards, and tutorials.

65. **Scholarly Impact & Altmetrics (Priority: Medium)** – Track metrics beyond downloads, such as social media mentions, citations of datasets, or usage in educational materials. Value: Shows impact of data and publications (e.g. dataset mentioned in news or Twitter).

- *Technical Feasibility*: Use altmetric APIs or custom scrapers for DOIs.
- *Complexity*: L
- *Risk*: Third-party API limits; Mitigation: cache data, update daily.
- *Data Model*: **AltMetric** (item_id, service, metric_value, last_checked) .
- *APIs*: GET /api/metrics?item={id} .
- *UI*: “Metrics” tab on item pages, charts of citations, tweets, readers count, etc.

66. **Offline & Mobile Support (Priority: Low)** – Provide a mobile-friendly interface and offline caching (e.g. store key data or maps locally for fieldwork). Value: Field researchers can use the app in remote locations without continuous internet.

- *Technical Feasibility*: Responsive design; Progressive Web App or native app.
- *Complexity*: L
- *Risk*: Complexity of syncing; Mitigation: focus on read-only caches (map tiles, select data subsets).
- *Data Model*: No new core data; supports offline caching.
- *APIs*: Standard APIs with offline support using ServiceWorkers.
- *UI*: Responsive UI, “Download for offline” option for pages/maps.

67. **Blockchain Ledger for Data Integrity (Priority: Low)** – (Proof-of-concept) Record dataset publication events and hashes on a blockchain for immutable provenance. Value: Ensures tamper-evidence and long-term trust.

- *Technical Feasibility*: Use Ethereum or Hyperledger to record hashes.
- *Complexity*: H (specialized knowledge, low familiarity)
- *Risk*: Cost (gas fees), obscurity; Mitigation: limit to critical datasets, or use consortium blockchain.

68. **Customisable Dashboards & Visual Analytics (Priority: Low)** – Allow users to create personal dashboards (widgets for datasets, charts, maps, news). Value: Researchers and educators can tailor PS63 to their interests.

- *Technical Feasibility*: Widget system (React components) and backend saving preferences.
- *Complexity*: M
- *Risk*: UI complexity; Mitigation: Provide templates.

69. **Multi-language AI Translation (Priority: Low)** – Use AI to translate content (papers' abstracts, glossaries, UI). Value: Non-English speakers access content (beyond static translations already planned).

- *Technical Feasibility*: Use services like Google Translate API or open models.
- *Complexity*: M
- *Risk*: Mistranslation; Mitigation: Show original alongside translation, mark uncertain terms.

Note: Priorities (H/M/L) reflect impact vs effort. High-priority items are near-term “must-haves” (e.g. federated search, AI recs, real-time data) that greatly enhance utility. Medium/low items are innovative or value-add features (e.g. 3D maps, blockchain) that can be phased in later.

Data Models, APIs, and UI Components by Feature

For each feature above, key elements would include:

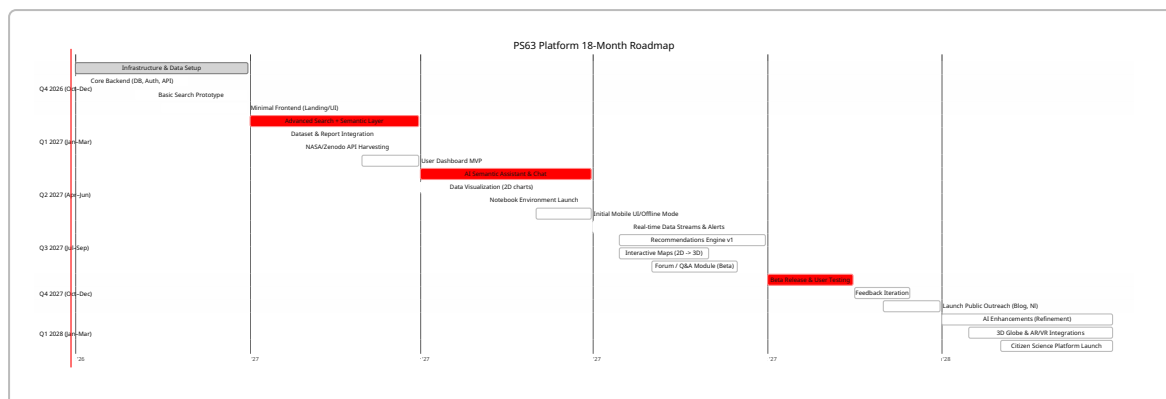
- **Real-time Streams** – *Model*: `SensorData(id, timestamp, lat, lon, sensor_type, value, quality)`. *API*: `/api/sensors/stream` for data ingest; `/api/sensors/query?start=&end=&bbox=`. *UI*: Live data dashboard (charts with play/pause), map overlays.
- **Federated Search** – *Model*: `ExternalRecord(id, source, title, url, summary, metadata)`. *API*: `/api/search?query=...` integrates multiple sources; `/api/records/{id}` fetch by source. *UI*: Unified search bar, results list with source icons (Zenodo, NASA, etc).
- **Notebooks/Workspaces** – *Model*: `AnalysisSession(id, user_id, name, created_at, status, last_active)`. *API*: `/api/workspaces` (create/list); `/api/workspaces/{id}/files` (list notebooks); `/api/workspaces/{id}/run` (execute). *UI*: “My Projects” page; Jupyter frontend embedded via iframe or redirect.
- **Recommendations** – *Model*: `UserInteraction(user_id, item_type, item_id, action, timestamp)`. *API*: `/api/users/{id}/recommendations`. *UI*: “For you” carousels on homepage and item pages.

- **3D Visualisations** – *Model:* Entities like `PointFeature(lat, lon, alt, data_link)`, `VectorLayer(geojson)`. *API:* `/api/3ddata?region=...`, `/api/expeditions/{id}/route3d`. *UI:* 3D globe view (Cesium canvas) with expedition tracks and data layers toggles.
- **Data Provenance** – *Model:* `Provenance(id, item_id, agent, activity, timestamp, description)`. *API:* `/api/provenance?item={id}` returning PROV-JSON. *UI:* Interactive graph (e.g. nodes for datasets, persons, transformations) via D3.
- **Metadata Curation** – *Model:* extend existing records with `suggested_tags` or `location_entities`. *API:* `/api/metadata/suggest?item={id}` returns suggestions. *UI:* “Review Metadata” page showing original vs AI-suggested fields to accept.
- **Q&A Forum** – *Model:* `Thread(id, item_id, title)`, `Post(id, thread_id, user_id, content, created_at)`. *API:* `/api/threads?item={id}`, `/api/threads/{id}/posts`. *UI:* Discussion panel with Markdown support, upvote buttons, user avatars.
- **Multi-modal Search** – *Model:* `MediaLabel(media_id, label, confidence)`, `VideoTranscript(video_id, start, text)`. *API:* `/api/search/images?q=object`, `/api/search/videos?q=word`. *UI:* “Search by Image” button; video player with transcript timeline highlight.
- **Citizen Science** – *Model:* `Task(id, description, media_ids)`, `Contribution(id, task_id, user_id, response)`. *API:* `/api/citizen/tasks`, `/api/citizen/tasks/{id}/submit`. *UI:* Gamified task interface (point scoring, visual tasks like tagging).
- **Impact Metrics** – *Model:* `Metric(item_id, metric_type, count, updated_at)` for altmetrics (Twitter, Mendeley, citations). *API:* `/api/metrics?item={id}`. *UI:* Metrics dashboard with charts (bar for citations per year, bubble for tweets).
- **Offline Support** – *Model:* N/A (feature of client caching). *API:* no change. *UI:* “Offline Mode” status, buttons to save page or map tiles.
- **Blockchain Integrity** – *Model:* `BlockRecord(item_id, tx_hash, timestamp)`. *API:* `/api/blockchain/verify?item={id}` returns stored hash. *UI:* “Verified on blockchain” badge with link to transaction.
- **Dashboards** – *Model:* `Dashboard(id, user_id, layout, widgets)`. *API:* `/api/dashboards`, `/api/dashboards/{id}`. *UI:* Drag-and-drop widget canvas (charts of user's data, maps, RSS).
- **Translation** – *Model:* `Translation(text_id, language, translated_text)`. *API:* `/api/translate?text={id}&lang=es`. *UI:* “Read in Spanish” button opens translated abstract.

Each feature's implementation will reuse and extend PS63's core models (User, Expedition, Dataset, etc.), linking via foreign keys. All new APIs follow REST conventions (JSON in/out), secured by existing auth roles. UI components (built in e.g. React or Vue) will be modular and fit into the PS63 design system (e.g. a Polaris or Material-UI theme).

12–18 Month Roadmap

The roadmap is phased by quarter to balance early wins and dependencies:



Milestones/Deliverables: By each quarter’s end: functional search, upload, visualization, and AI modules as above.

Dependencies: Foundation in Q4 (auth, DB, basic UI). Search/semantic built on core models. AI features rely on trained models (start collecting data early).

Resources (per quarter estimate):

- Dev (backend/frontend): 3–5 FTEs working concurrently (increasing Q1–Q2), tapering later.
- Data (ingest, curation): 1–2 FTEs (the data engineer team).
- ML/AI: 1–2 specialists (for AI/semantic assistants, embeddings, etc., intensifying Q2-Q3).
- QA: 1–2 across the project (testing each milestone).
- Infrastructure: devops support for cloud infra (AWS/GCP) – initial heavy lift in Q4, then maintenance.

Competitor/Benchmark Comparison

Platform	Focus/Users	Key Features (present)	PS63 Gaps & Opportunities	Sources
Zenodo	General science data (CERN/ OpenAIRE)	Free repository for datasets & code; DOIs for every upload; versioning; usage stats; GitHub integration [8†L281-L289] . Open/ restricted access.	Lacks domain specialization (not polar-specific) and advanced search – PS63 can add geospatial context, AI tools.	[8†L281-L289]
Dryad	Life-science data (publishers)	Nonprofit curated data; DOI for each dataset; strong curation and compliance checks [14†L25-L33] ; ORCID integration.	Focuses on biodiversity; no mapping or AI search. PS63 can target polar science and add mapping, semantic tools and community features.	[14†L25-L33]

Platform	Focus/Users	Key Features (present)	PS63 Gaps & Opportunities	Sources
PANGAEA	Earth & environmental science	Georeferenced dataset repository (hosted by AWI) with advanced search, GIS integration and data projects 【24†L6-L10】 . APIs for data.	Lacks social/Q&A, AI summarization, semantic linking beyond classic metadata. PS63 can layer AI discovery, knowledge graph and outreach.	【24†L6-L10】 【24†L163-L166】
NOAA Data Portals	Oceans, atmosphere (US gov)	Massive climate/ ocean datasets; map layers; download tools (e.g. NCEI), specialized indices. Often fragmented by sub-agency.	No integrated researcher profiles or semantic AI features. PS63 can unify polar data from NOAA and provide researcher-centric view.	(public domain knowledge)
NASA Earthdata	Earth observation (NASA)	Gigantic satellite/ gridded data catalogue; Earthdata Search for filtering data; cloud access (AWS S3) 【83†L99-L104】 ; APIs.	Users need heavy domain knowledge; lacks community and data synthesis tools. PS63 can add easy interpretation (AI summaries, VR).	【83†L99-L104】
ResearchGate	Scientific social network	Researcher profiles (26M+ users); publication sharing; Q&A forum; impact stats (reads, citations); follows. No raw data repository.	Strong in social networking but weak on data hosting. PS63 can combine data repository + social features (citizen science, team spaces).	(industry knowledge)

Benchmark Gaps: None of the above integrate **semantic AI search, knowledge graphs, real-time data streams, interactive polar mapping, or a combined researcher profile ↔ dataset ecosystem**. PS63 can exploit these by offering a domain-tailored, AI-enhanced portal (e.g. tie expedition → researcher → data → publications as a connected knowledge graph) **【89†L236-L244】** .

Recommended Technology Stack

- **Frontend:** Modern JavaScript framework (e.g. **React/Next.js** or **Vue/Nuxt**). Supports dynamic components (maps, charts) and PWA for offline mode. Rationale: large ecosystem, good for data dashboards and UI flexibility. Cost/scale: open-source, easily containerized.
- **Backend:** **Python/Django** or **Node.js/Express** microservices. Django offers rapid development, ORM for complex models (or use Node for JS full-stack). Rationale: strong libraries (Django Rest Framework) for research data.

- **Database:** Hybrid. **PostgreSQL** for relational data (users, metadata); **Elasticsearch** (or Opensearch) for fast search, semantic KNN search (vector indexes); **Neo4j** or **OrientDB** (Graph DB) for knowledge graph/lineage. Rationale: PG for transactions, ES for full-text/vector search [\[65†L122-L130\]](#) , Graph DB to store and query relationships. Cost: all have open-source editions; scale on cloud DB services or self-hosted.
- **Search/Semantic:** **Elasticsearch** with a vector plugin (or **Pinecone/Qdrant** vector DB) to index text and embeddings for semantic search. Use pre-trained models (e.g. Sentence-BERT) for embeddings. Rationale: Combines keyword + semantic search (hybrid) [\[65†L122-L130\]](#) .
- **AI/ML Infra:** GPU-enabled servers or cloud ML services. Use frameworks like **Hugging Face Transformers** or **OpenAI API** for summarization/translation. Fine-tune domain models if needed. Deploy ML models behind REST endpoints (e.g. FastAPI). Rationale: leverage existing LLMs rather than train from scratch.
- **Mapping & Visualization:** **Leaflet** or **OpenLayers** for 2D maps; **CesiumJS** for 3D globe. **D3.js** or **Plotly** for interactive charts. Rationale: these are proven libraries for geospatial and scientific viz.
- **Hosting/Infra:** Cloud (AWS/GCP/Azure) with container orchestration (Kubernetes or Docker Swarm) for scalability. Use **S3** (or equivalent) for object storage (datasets, media). Use **AWS/GCP** managed databases and Elasticsearch for reliability. Rationale: supports large datasets and spikes in AI compute. Budget-neutral: open-source stack runs well on mid-tier instances.
- **CI/CD:** GitHub Actions or GitLab CI to automate testing, linting, deployment. Container registry (Docker Hub or ECR). Rationale: standard DevOps.

(Cost/scale note: using managed services (e.g. AWS RDS, Elasticsearch Service) reduces ops overhead at modest cost. For semantic search and AI models, monitor usage—open-source models can run on GPU instances or use serverless inference to control costs.)

Privacy, Security & Governance Checklist

To comply with best practices and build trust:

- **Authentication & RBAC:** Enforce strong login (e.g. OAuth2, password policies). Role-Based Access Control as in feature #31 (Admin/Researcher/Reviewer etc) for data upload and visibility.
- **Encryption:** All data in transit via HTTPS/TLS; sensitive PII (if any) encrypted at rest.
- **Data Privacy:** Collect minimal personal data. For researcher profiles, allow opting out of public display. Comply with relevant laws (GDPR) if storing EU user data.
- **Audit Logs:** Record actions (uploads, approvals, edits) for accountability. E.g. log reviewer actions for traceability.
- **Data Governance:** Adopt FAIR principles [\[89†L236-L244\]](#) – require clear metadata and licenses. Metadata should include creator, date, license (CC0/CC-BY encouraged).
- **Source Verification Workflow (for AI answers):** All AI-generated summaries or answers link back to actual PS63 content (reports, papers). Process: user query → retrieve relevant source documents (from PS63) → run LLM on those docs only → provide answer with citations to the original sources. Maintain chain of custody so AI never hallucinates unsupported claims.
- **Vetting & Moderation:** Human-in-the-loop review for sensitive content: e.g. vet AI-generated outreach posts, forum posts flagged by users.
- **Quality Checks:** Use checksum/DOI mechanisms (per FAIR) for data integrity; flag outdated or superseded data.
- **Backup & Recovery:** Regular backups of DB and file storage; disaster recovery plan.
- **Open Licensing:** Encourage open licenses (CC-BY, CC0) and clearly display them on each resource.

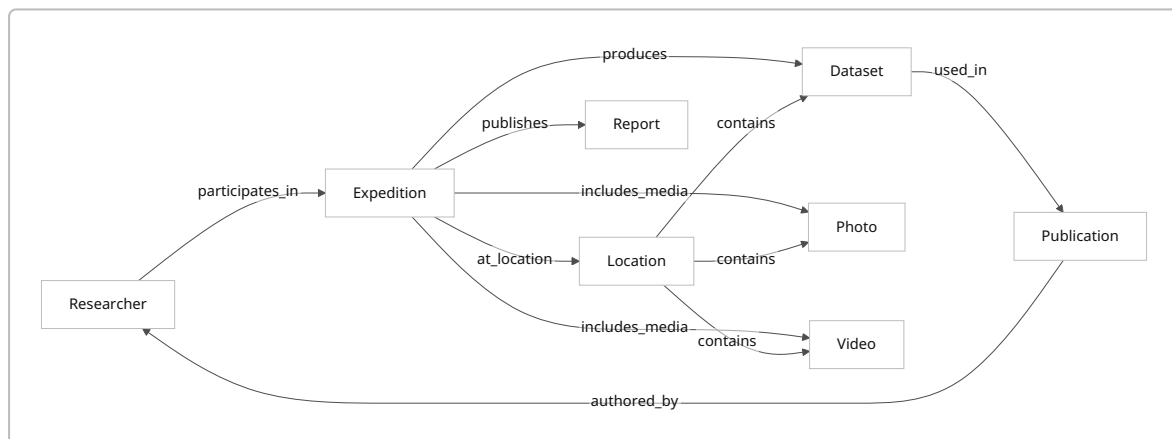
- **Reviewer Verification:** All submitted research must link to verifiable author profiles (e.g. ORCID). Reviewer approval is tracked (per feature #17).

Sample Developer Prompts

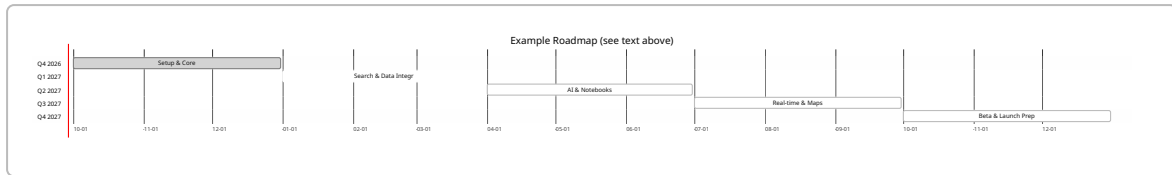
Product Brief (High-level): “PS63 is a unified polar science portal. Researchers register (using ORCID), create “expeditions” (with team members and geospatial data), and upload reports, datasets, and media. The platform assigns DOIs to all outputs. Users can search by keywords or semantically (AI-powered), and explore via maps and timelines. AI assistants can summarise reports or explain them simply. All content (data, papers, photos) is interlinked: clicking a station shows related expeditions, publications and personnel. Students get an educational view. The system requires role-based review and approval before public release. Provide features and APIs to support: user auth, expedition management, dataset upload/preview, semantic search, AI Q&A, interactive maps, social features (bookmarks, sharing), and admin dashboard (analytics, pending reviews).”

Technical Implementation Prompt: “Design a scalable PS63 web application with these components: a relational DB (PostgreSQL) for core entities (Users, Researchers, Expeditions, Datasets, Publications, Media); a search index (Elasticsearch or vector DB) for full-text and semantic search; a Graph DB for the knowledge graph. Define JSON schemas: e.g. `Expedition(id, name, start_date, end_date, description, lead_researcher_id, locations[], members[])`, `Dataset(id, expedition_id, title, variables[], spatial_extent, temporal_extent, file_refs[])`. Provide RESTful API endpoints: e.g. `POST /api/expeditions`, `GET /api/search?query=`, `POST /api/ai/ask` (body: user question + doc ids) returning a JSON answer with `sources` array. Outline CI/CD pipeline: use GitHub Actions to run lint, test suites (including data upload tests), and deploy Docker containers to AWS ECS or GCP Cloud Run on push to main. Include unit, integration and end-to-end tests for upload, search, and AI features. Use environment config to toggle prototype (mock AI) vs production (real AI model). Ensure data validation and error handling in the API (400/500 codes).”

Illustrative Diagrams



Mermaid graph: Entities in PS63 and their relationships (a simplified knowledge graph schema).



Mermaid Gantt chart: High-level PS63 development phases by quarter.

Sources: Descriptions of existing platforms are drawn from their official sites [8†L281-L289] [14†L25-L33] [24†L6-L10] and the FAIR data literature [89†L236-L244] . Semantic search concepts are from information retrieval research [65†L122-L130] . These inform the above proposals.